

Proportion & the Rule of Three

When two ratios are equal we have a proportion. Cross multiplication turns it into a quick way to find an unknown — the ancient Rule of Three.

Proportion & cross multiplication

The idea: $a : b :: c : d$ means the two ratios are equal. Then the **cross products are equal:** $a \times d = b \times c$.

$$a : b :: c : d \iff ad = bc$$

The Rule of Three (Trairāśika)

When three quantities of a proportion are known and one is unknown, set up $a : b :: c : x$ and cross-multiply to find x .

$$6 : 10 :: 18 : x \rightarrow 6 \times x = 10 \times 18 \rightarrow x = 180 \div 6 = 30$$

Keep the units consistent

Both ratios must use the **same units**. A car going 90 km in 150 minutes — to find the distance in 4 hours, first change 4 hours to 240 minutes.

$$150 : 90 :: 240 : x \rightarrow 150x = 90 \times 240 \rightarrow x = 144 \text{ km}$$

Quick proportion check: to test whether $a : b :: c : d$, just compare the cross products ad and bc . If they're equal, it is a true proportion.

Prerequisite refresher: "Cross products" are the diagonal products $a \times d$ and $b \times c$ of $a : b :: c : d$.

Where marks slip away

Trap 1 — Mismatched units. Convert everything to one unit before forming the proportion (minutes with minutes, grams with grams).

Trap 2 — Cross the right terms. For $a : b :: c : d$, multiply $a \times d$ and $b \times c$ — the diagonals, not $a \times c$.

Trap 3 — Set up the order carefully. Keep "first quantity" with "first quantity" on both sides of the $::$.

Trap 4 — Not everything is a proportion. Faster speed $\rightarrow$ less time (inverse), so a simple Rule of Three doesn't apply when one goes up as the other goes down.

Slip 5 — State the unit of the answer (km, kg, spoons...), not just the number.

Model answers — write them like this

Example A. Is $4 : 7 :: 12 : 21$ a true proportion?

Step 1: Cross products: $4 \times 21 = 84$ and $7 \times 12 = 84$.

Step 2: They are equal, so the proportion is true.

$\therefore$ Yes, it is a true proportion

Example B. Solve for x: $6 : 10 :: 18 : x$.

Step 1: Cross multiply: $6 \times x = 10 \times 18 = 180$.

Step 2: $x = 180 \div 6 = 30$.

$\therefore x = 30$

Example C. A car travels 90 km in 150 minutes. How far will it go in 4 hours at the same speed?

Step 1: Make units match: 4 hours = 240 minutes.

Step 2: $150 : 90 :: 240 : x$, so $150x = 90 \times 240 = 21600$.

Step 3: $x = 21600 \div 150 = 144$.

$\therefore 144 \text{ km}$

Example D. For 120 students a cook makes 15 kg of rice. How much rice for 80 students?

Step 1: $120 : 15 :: 80 : x$.

Step 2: $120x = 15 \times 80 = 1200$, so $x = 10$.

$\therefore 10 \text{ kg}$

Next: [try the worksheet](#), then [check the answer key](#).